General Use SOP

Cryogenic Liquids and Dry Ice

Process or Experiment Description

This standard operating procedure (SOP) provides general guidance on working safely with lab-scale quantities of cryogenic liquids and dry ice. Cryogenic liquids that are also reactive (e.g., liquid ethane, liquid oxygen) require additional precautions beyond those in the relevant General Use SOPs (e.g., flammable and combustible liquids for liquid ethane, highly reactive/unstable materials for liquid oxygen), and labs should develop specific SOPs for working with them. For questions about the applicability of any item in this SOP, contact your Principal Investigator/Supervisor or EH&S at 650-723-0448.

For work with large quantities of cryogenic liquids or dry ice, complex procedures, or otherwise particularly hazardous procedures involving cryogenic liquids or dry ice, EH&S recommends completing a risk assessment, reviewing other EH&S guidance (such as guidance on scale-up), looking for SOPs on similar procedures in the SOP Library, and/or writing a dedicated SOP and submitting it for EH&S review.

Hazards

Cryogenic liquids are materials with extremely low boiling points (i.e., below -150 °F/-101 °C). Common examples of cryogenic liquids include liquid nitrogen, liquid helium, and liquid argon. Dry ice, the solid state of carbon dioxide, has a surface temperature of -109.3 °F/-78.5 °C. Cryogens' low temperatures can cause cryogenic burns and frostbite. Both cryogenic liquids and dry ice undergo substantial volume expansion upon evaporation or sublimation, potentially leading to an oxygen deficient atmosphere. This expansion can also cause pressure buildup, which may cause an explosion if a container lacks sufficient pressure relief. Cryogenic liquids and dry ice can make some materials brittle during prolonged exposure. Some cryogenic liquids can also pose additional hazards, including toxicity and flammability (i.e., liquid carbon monoxide).

Control of Hazards

General

- Only work with cryogenic liquids in well-ventilated areas like wet labs to avoid localized oxygen depletion or the buildup of flammable or toxic gas.
- Use the smallest feasible quantity.
- Handle objects in contact with cryogenic liquids with tongs or cryogenic gloves.
- Transfers or pouring of cryogenic liquids should be done carefully to avoid splashing.
- Containers and systems containing cryogenic liquids should have pressure relief mechanisms.
- Cryogenic liquid cylinders and other containers (such as Dewar flasks) should be filled to no more than 80% of their capacity.
- Cryogenic liquid/dry ice baths should be open to the atmosphere to avoid pressure build up. Do not put cryogenic liquids or dry ice into closed, sealed containers without pressure relief. Ensure evolved gas can be released.
- Cryogenic liquids with boiling points below that of liquid oxygen (-297.3 °F/-183 °C) can condense oxygen in them under certain conditions. Liquid oxygen is a pale blue liquid that is extremely reactive and can cause combustible materials to spontaneously ignite, and will explode when in contact with flammable liquids.
- When retrieving dry ice from a storage box, do not lean over or into the box, as it will be full of concentrated gaseous CO2 which can rapidly render you unconscious. Use a long-handled scoop to remove the dry ice.
- When thawing cryotubes, place the cryotube in a heavy-walled container (e.g., a desiccator, cardboard box) or behind a safety shield to protect against shattering.
- Shield or wrap fiber tape around glass dewars to minimize flying glass and fragments should an explosion occur. Note: plastic mesh will not stop small glass fragments.

Engineering/Ventilation Controls

Use cryogenic liquids in well-ventilated areas such as wet labs. If the process doesn’t allow working with cryogenic liquids in a well-ventilated area, contact Environmental Health & Safety at 650-723-0448 to determine whether an oxygen-deficiency monitor is necessary.

Personal Protective Equipment

In addition to proper street clothing (long pants or equivalent that covers legs and ankles, and close-toed non-perforated shoes that completely cover the feet), wear the following Personal Protective Equipment (PPE):

- Safety glasses (If splash potential exists, wear goggles and a face shield instead)
    - Cryogenic liquids bubble, spit, and splash easily. Wear goggles and a face shield when working with moderate quantities of cryogenic liquid.
- Lab coat
- Insulated cryogenic gloves
- Optional: cryogen-resistant apron (Recommended when pouring large quantities of cryogenic liquids or retrieving samples from cryogenic storage)
- Optional: cryogen-resistant spats (Recommended when pouring large quantities of cryogenic liquids or retrieving samples from cryogenic storage)

Special Handling Procedures and Storage Requirements

- Store cryogenic liquid dewars in well-ventilated areas. Storage in unventilated closets, environmental rooms, and stairwells is prohibited.
- Anchor/tether large dewars to a wall.
- Store flammable cryogenic liquids and liquid oxygen away from combustible materials and sources of ignition.
- Follow all substance-specific storage guidance provided in SDS documentation.
- If vessels containing cryogenic liquids or dry ice become frozen shut or covered with ice, contact the manufacturer for guidance on how to thaw them. Excessive heating can cause pressure to build within vessels sealed by ice.

Emergency Procedures

All incidents should be reported to EH&S with an incident report form.

Spill

Refer to EH&S' guidance on hazardous materials incidents for general procedures for incident response, and call 650-725-9999 for assistance with spill cleanup.

Do not attempt to clean up any spill of cryogenic liquid. If a small spill occurs, block off the area and allow the material to evaporate. If a large spill or dewar leak occurs, immediately evacuate the area and call 650-725- 9999 for emergency assistance.

Exposure

If immediate medical attention is required, call 911. Remove any clothing that is trapping or pooling cryogenic liquid or dry ice; otherwise loosen but do not remove clothing in the affected area, and IMMEDIATELY flush contaminated skin with room-temperature water for at least 15 minutes following any skin contact. DO NOT flush with hot water. Use a safety shower for any skin exposures to the torso, legs, or feet. Do not rub or massage exposed areas.

For eye exposures, IMMEDIATELY flush eyes with water for at least 15 minutes using an eyewash station.

If medical attention is needed, provide the SDS(s) of the chemical(s) to aid medical staff in proper diagnosis and treatment.

For minor exposures, personnel can contact the Stanford University Occupational Health Center at 650-725-5308. For serious exposures, personnel should go to the Stanford Hospital Emergency Department.

Those who work with hazardous chemicals are entitled to receive medical attention/consultation when:

- A spill, leak, explosion or other occurrence results in a hazardous exposure (potential overexposure).
- Symptoms or signs of exposure to a hazardous chemical develop.

Liquid Oxygen Formation

If liquid oxygen condenses in a cryogenic liquid bath or a vessel cooled by cryogenic liquid, immediately stop work and notify others in the area. Remove the external cryogenic liquid bath. Remove combustible materials from the immediate vicinity, then evacuate the area and call 650-725-9999. If there is a spill of liquid oxygen, evacuate the area immediately and call 650-725-9999.

Waste Disposal

Coordinate with the appropriate vendor for return of cryogenic liquid cylinders or dry ice boxes. If you otherwise need assistance, contact EH&S.

Minimum Training Requirements

- General Safety & Emergency Preparedness (EHS-4200)
- Chemical Safety for Laboratories (EHS-1900)
- Compressed Gas Safety (EHS-2200)
- Cryogenic Liquids and Dry Ice Safety (EHS-2480)
- If shipping samples with dry ice: Shipping Dangerous Biological Goods or Dry Ice (EHS-PROG-2700)
- Laboratory-specific training

Approval Required

Consult with your PI regarding the need for prior approval. Laboratory personnel shall seek and the PI must provide prior approval of any chemical usage involving Restricted Chemicals or High Risk Procedures.

Designated Area

N/A